

Program 7: Deploy a Web Application to Kubernetes using Minikube

Project Structure

DEVOPSLAB

|--program7

|--nginx-deployment.yaml

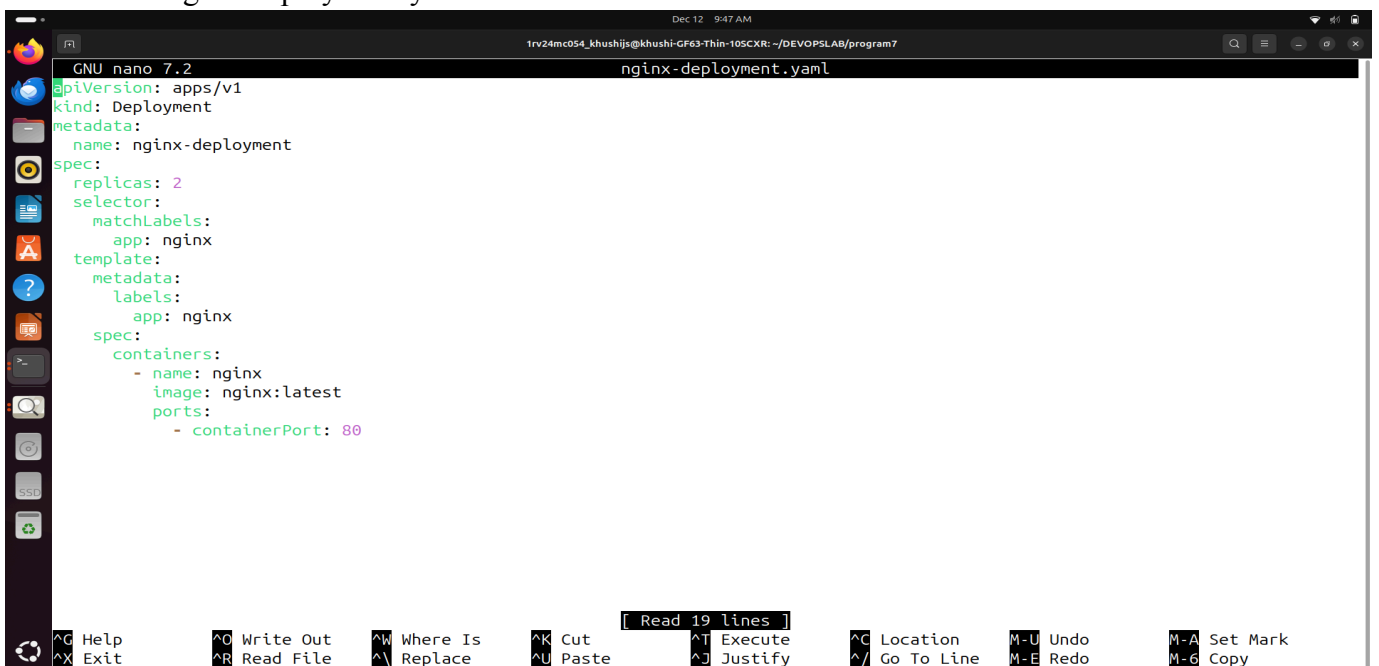
Step 1: Start the minikube

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~$ cd DEVOPSLAB
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB$ mkdir program7
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB$ cd program7
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program7$ minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.48 ...
🔄 Updating the running docker "minikube" container ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: default-storageclass, storage-provisioner
🏠 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

Step 2: Check the status of minikube

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program7$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 3: Create deployment file nano nginx-deployment.yaml



```
GNU nano 7.2
nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Step 4: Apply the nginx-deployment.yaml

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
```

Step 5: Check status

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program7$ kubectl get deployments
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
nginx-deployment	2/2	2	2	94s

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program7$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-6f9664446b-q5vbf	1/1	Running	0	103s
nginx-deployment-6f9664446b-x84kq	1/1	Running	0	104s

Step 6: Expose the deployment as a Service

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program7$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
```

Step 7: Check service

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program7$ kubectl get services
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	12h
nginx-deployment	NodePort	10.102.158.209	<none>	80:32600/TCP	22s

Step 8: Access the Web App

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program7$ kubectl get services
NAME          TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes    ClusterIP     10.96.0.1     <none>          443/TCP          12h
nginx-deployment NodePort      10.102.158.209 <none>          80:32600/TCP     22s
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program7$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:32600

Opening service default/nginx-deployment in default browser...



Step 9: Clean-up process

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program7$ kubectl delete pods --all
pod "nginx-deployment-6f9664446b-q5vbf" deleted from default namespace
pod "nginx-deployment-6f9664446b-x84kq" deleted from default namespace
pod "nodejs-pod" deleted from default namespace
```

Step 10: Delete all services

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program7$ kubectl delete svc --all
service "kubernetes" deleted from default namespace
service "nginx-deployment" deleted from default namespace
```

Step 11: Delete all deployments

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program7$ kubectl delete deployments --all
deployment.apps "nginx-deployment" deleted from default namespace
```